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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Fifth Semester B.Tech (CBCS) Degree Examinations

December 2025

BT1652-1 – BIOETHICS & BIOSAFETY
 (For BT)

Duration: 2 Hours

Max. Marks: 50

Note: Descriptive Answer Questions: Answer **Three full** questions choosing **One full** question each from **Unit – I, II & III.**

Unit – I		Marks	BT*	CO*	PO*
1.	a) Explain how scientific discoveries drive technological advancements and impact society.	10	L*3	1	1,2,6
	b) Define bioethics committee. List out the composition of ethics committee.	5	L2	2	6,8
2.	a) Comment on the bioethical principles which influences biomedical research.	8	L2	1	1,2,6
	b) Evaluate how public perception influences the acceptance of biotechnology products.	7	L3	2	1,2,6
Unit-II					
3.	a) Give a brief description on the use of animals in research and testing. Add a note on the alternatives for animals in research.	10	L2	3	1,6,8
	b) Write a brief note on the ethical issues in organ transplantation.	5	L2	3	1,6,8
4.	a) Analyze the ethical considerations with response to bioterrorism threats.	8	L3	3	6,8
	b) Discuss the ethical issues associated with prenatal diagnosis.	7	L3	3	1,6,8
Unit-III					
5.	a) Discuss the four levels of biosafety and their design, containment and safety requirements.	10	L2	4	1,2,5,8
	b) Briefly outline the goals and categories of good manufacturing practices.	10	L2	5	1,6,8
6.	a) Briefly discuss on the process of risk management.	10	L2	4	1,2,6,8
	b) Explain the objectives and key provisions of the Biological Diversity Act, 2002.	10	L3	5	1,6,8

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
